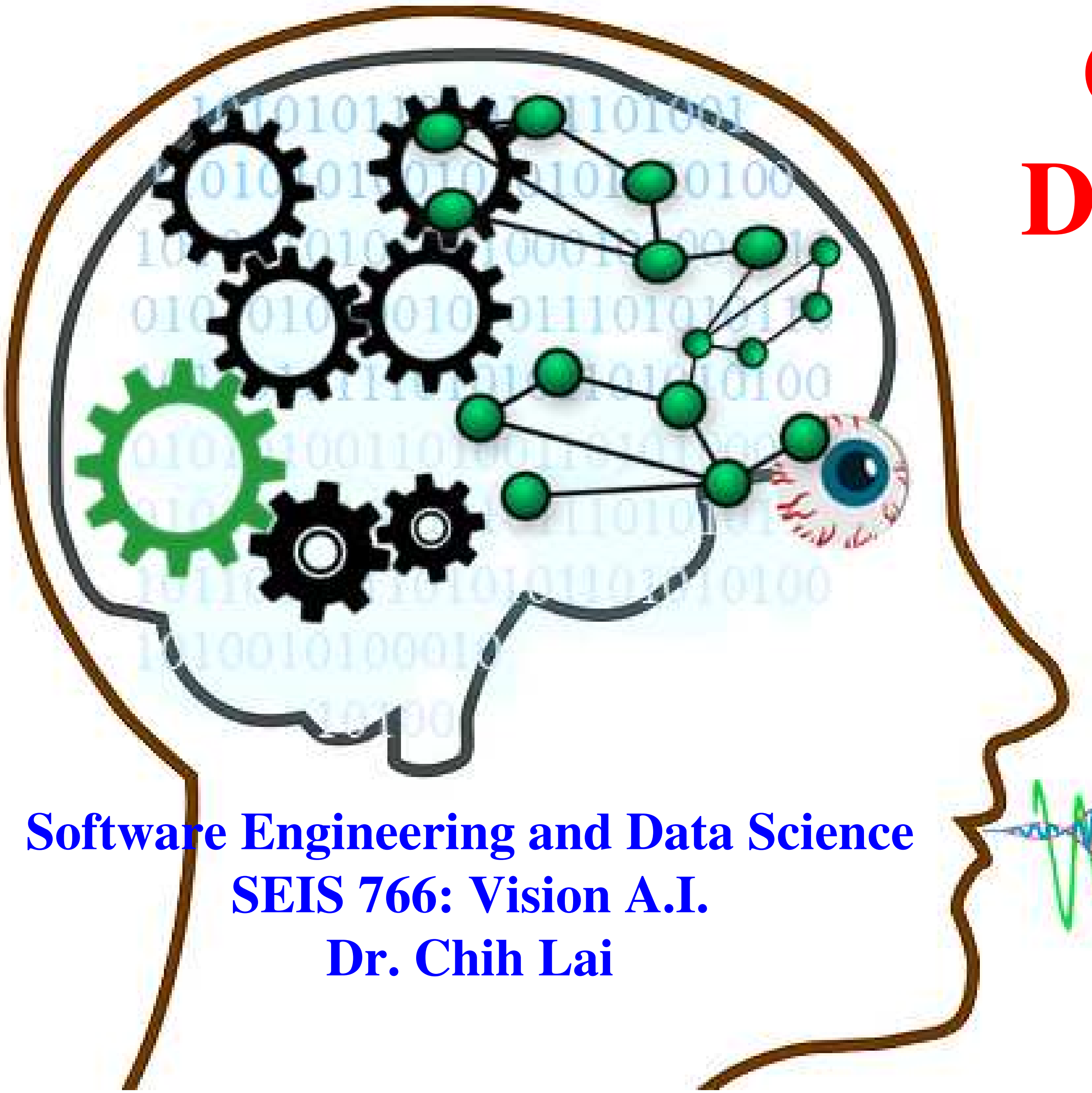
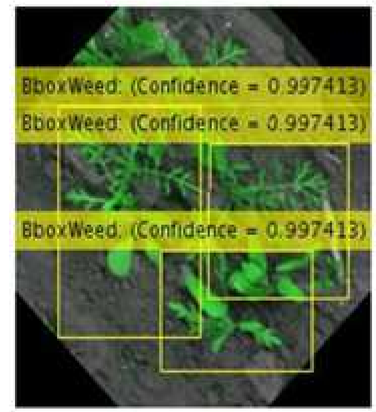
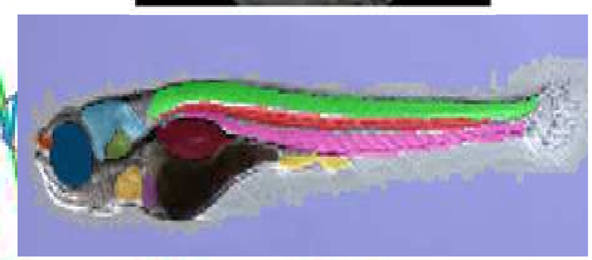
Object Detection

RCNN







Software Engineering and Data Science

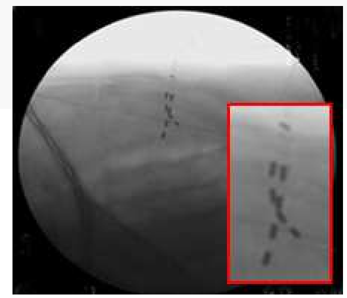
SEIS 766: Vision A.I.

Dr. Chih Lai

Classifying ***Whole*** Images, Enough???

- https://www.youtube.com/watch?v=nYprPrCTwrA



- Any problem???
  - **LOTs** of important apps **MUST** begin with object identification (detection)

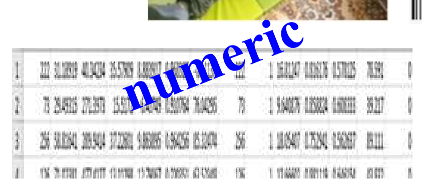
Just classify abnormal/normal organs??? **WHERE** are the organs… & **WHERE** are **abnormalities**???

Where is the leg, where is the knee, knee angle, leg/knee motion…

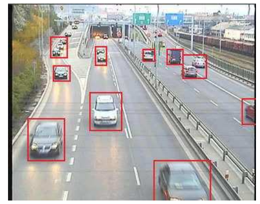
Where are the cars, distance to the car, counting cars, counting people

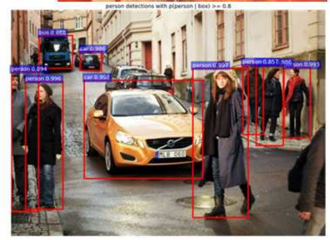
Where are devices, shape / angle / depth

Find similar business records, find similar patient histories

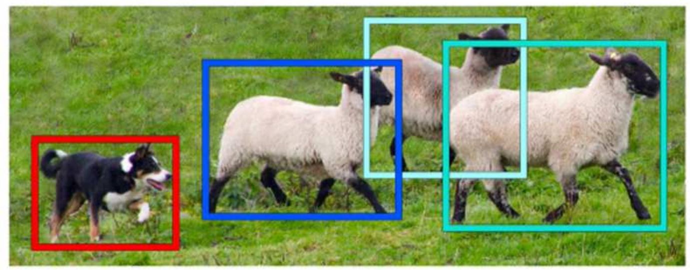
Different Types of Object Allocation

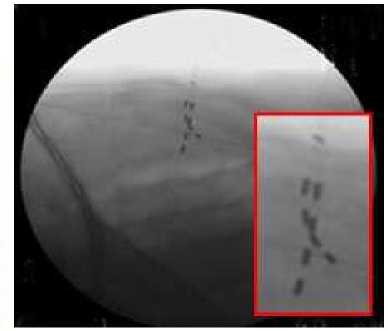




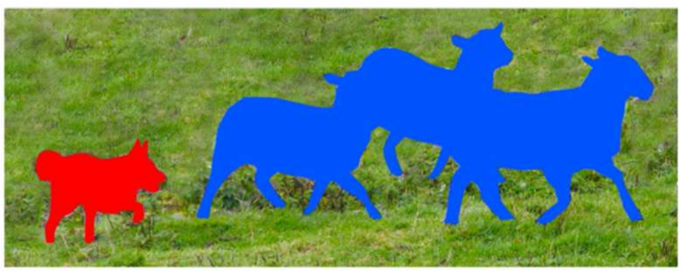
- RCNN
- Fast RCNN
- Faster RCNN
- Yolo
- SSD

Image Recognition

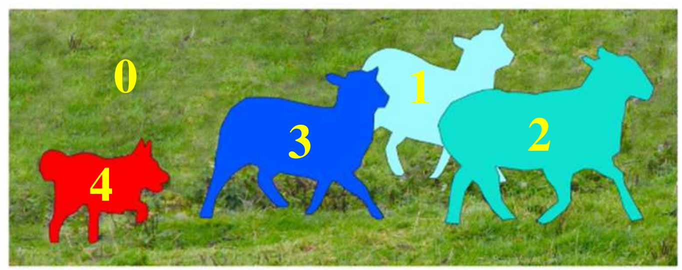




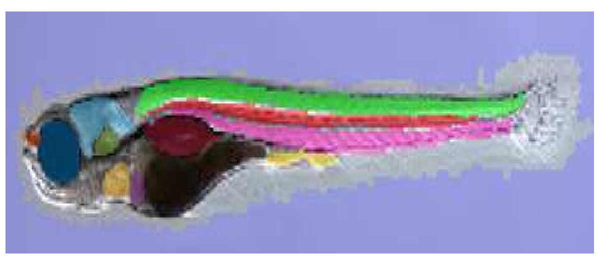
Object Detection

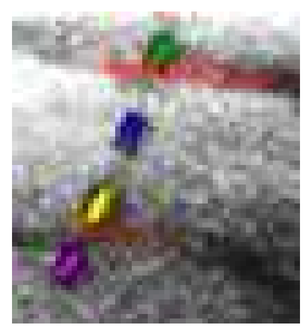


Semantic Segmentation



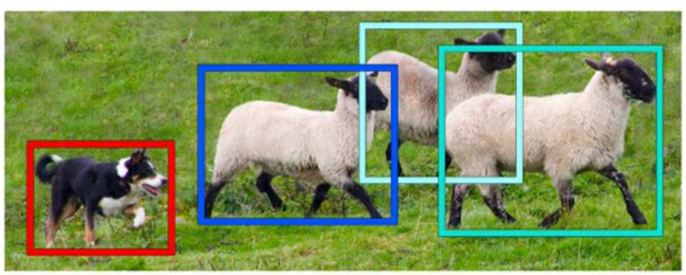
Instance Segmentation



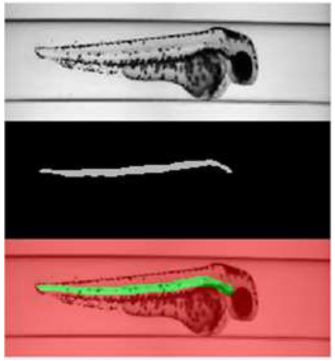


Outline

- √Object Detection, Fast RCNN
- Object Detection, Semantic Segmentation, Labeling, Applications
- Encode + Decode CNN, Transposed Convolution, Pre-Trained CNN
- Quality Evaluation of Object Detection
- Identifying Morphological Changes in Organs w/ Occlusion Map



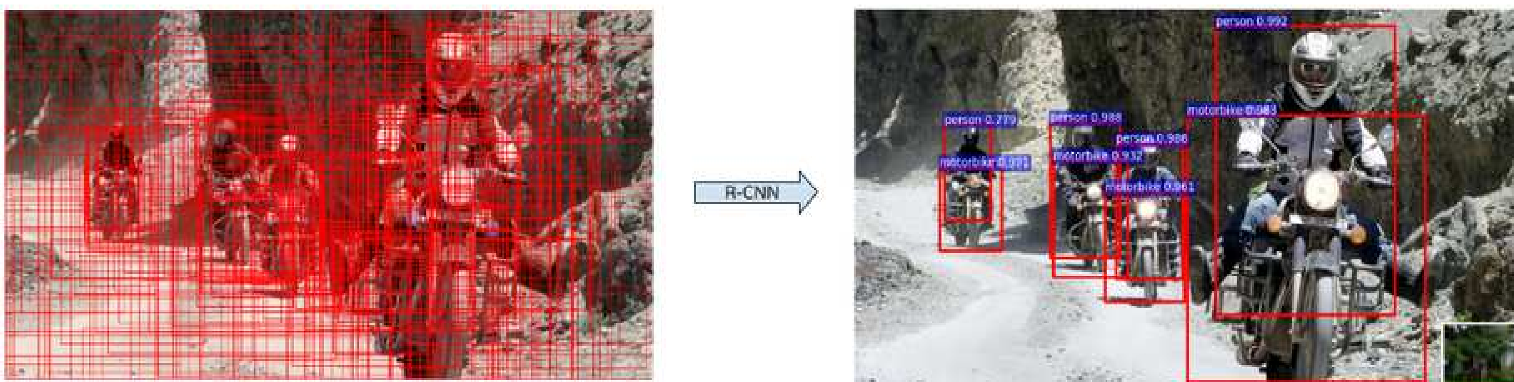
Object Detection



Object Detection in Images– R-CNN (**Regions** w/ CNN)

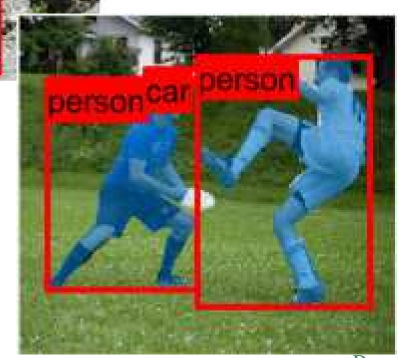
- Classify **EVERY** region using a sliding window over each image?
  - Use a CNN to classify **EVERY** sub-image (large amount!!) one at a time➔ time-consuming
  - Also, what should be the window size?

- R-CNN only processes regions that are ***likely*** to contain an object
  - Need **1.** ***Region Proposal*** (guessing regions) + **2.** Region Classification
    - Try to reduce the computational cost of brute-force CNNs
  - Alternatives: Fast RCNN, Faster RCNN



- ***Mask RCNN*** find bounding box **THEN find pixel regions**

(2017)



Pre-trained **cifar10Net** and Sample Training Images

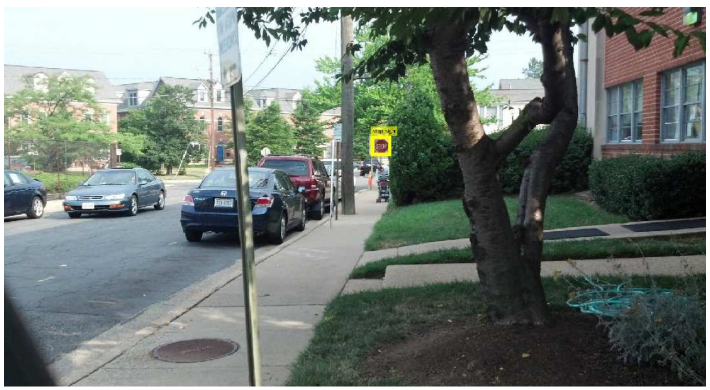
- **CIFAR-10 dataset** has **10 labels**: airplane, automobile, bird, cat, deer, dog, rog, horse, ship, truck.
  - with 50,000 images

- Since RCNN is basically classification on **sub-rectangles**, you can use many pre-trained classification CNN
  - Just need fine-tune training on the new label *Y* (on proposed subregions)
  - Divide images to subregions yourself….
  - Or, let pre-trained RCNN to find **sub-regions** for you

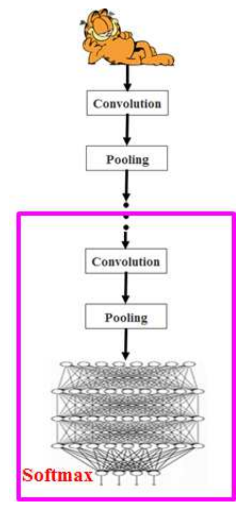
Transfer Learning with Fine-Tuning for The Specific Task

- Fine-tune '**cifar10Net**' for **<u>another</u>** task w/ much smaller training set
  - i.e. Fine-tune **cifar10Net** for ***stop sign detection*** w/ **<u>just 41</u>** training images
  - 41 images not enough to train a new CNN, but enough for **xfer learning**
- **cifar10Net Keras program:** https://www.kaggle.com/code/ektasharma/simple-cifar10-cnn-keras-code-with-88-accuracy

https://www.mathworks.com/help/vision/examples/object-detection-using-deep-learning.html?requestedDomain=www.mathworks.com



load('rcnnStopSigns.mat', 'cifar10Net')



Pre-Trained **RCNN**, **cifar10Net**

1	'imageinput'	Image Input	32x32x3 images with 'zerocenter' normalization
2	'conv'	Convolution	32 5x5x3 convolutions with stride [1 1] and padding [2 2]
3	'relu'	ReLU	ReLU
4	'maxpool'	Max Pooling	3x3 max pooling with stride [2 2] and padding [0 0]
5	'conv_1'	Convolution	32 5x5x32 convolutions with stride [1 1] and padding [2 2]
6	'relu_1'	ReLU	ReLU
7	'maxpool_1'	Max Pooling	3x3 max pooling with stride [2 2] and padding [0 0]
8	'conv_2'	Convolution	64 5x5x32 convolutions with stride [1 1] and padding [2 2]
9	'relu_2'	ReLU	ReLU
10	'maxpool_2'	Max Pooling	3x3 max pooling with stride [2 2] and padding [0 0]
11	'fc'	Fully Connected	64 fully connected layer
12	'relu_3'	ReLU	ReLU
13	'fc_rcnn'	Fully Connected	2 fully connected layer
14	'softmax'	Softmax	softmax
15	'classoutput'	Classification Output	cross-entropy with classes 'stopSign' and 'Background'

Classifying Each Box


```
load('rcnnStopSigns.mat', 'cifar10Net')

% Run the network on the test set.
YTest = classify(cifar10Net, testImages);
% Calculate the accuracy.
accuracy = sum(YTest == testLabels) / numel(testLabels)
```

https://www.mathworks.com/help/vision/examples/object-detection-using-deep-learning.html?requestedDomain=www.mathworks.com

- Evaluate RCNN w/ test data on classification accuracy based on **BOXes**
  - **Overlapping %** of **bounding boxes** **between *target box* and *predicted boxes***

Prepare Data *Y* = Rectangles for Fine-Tuning

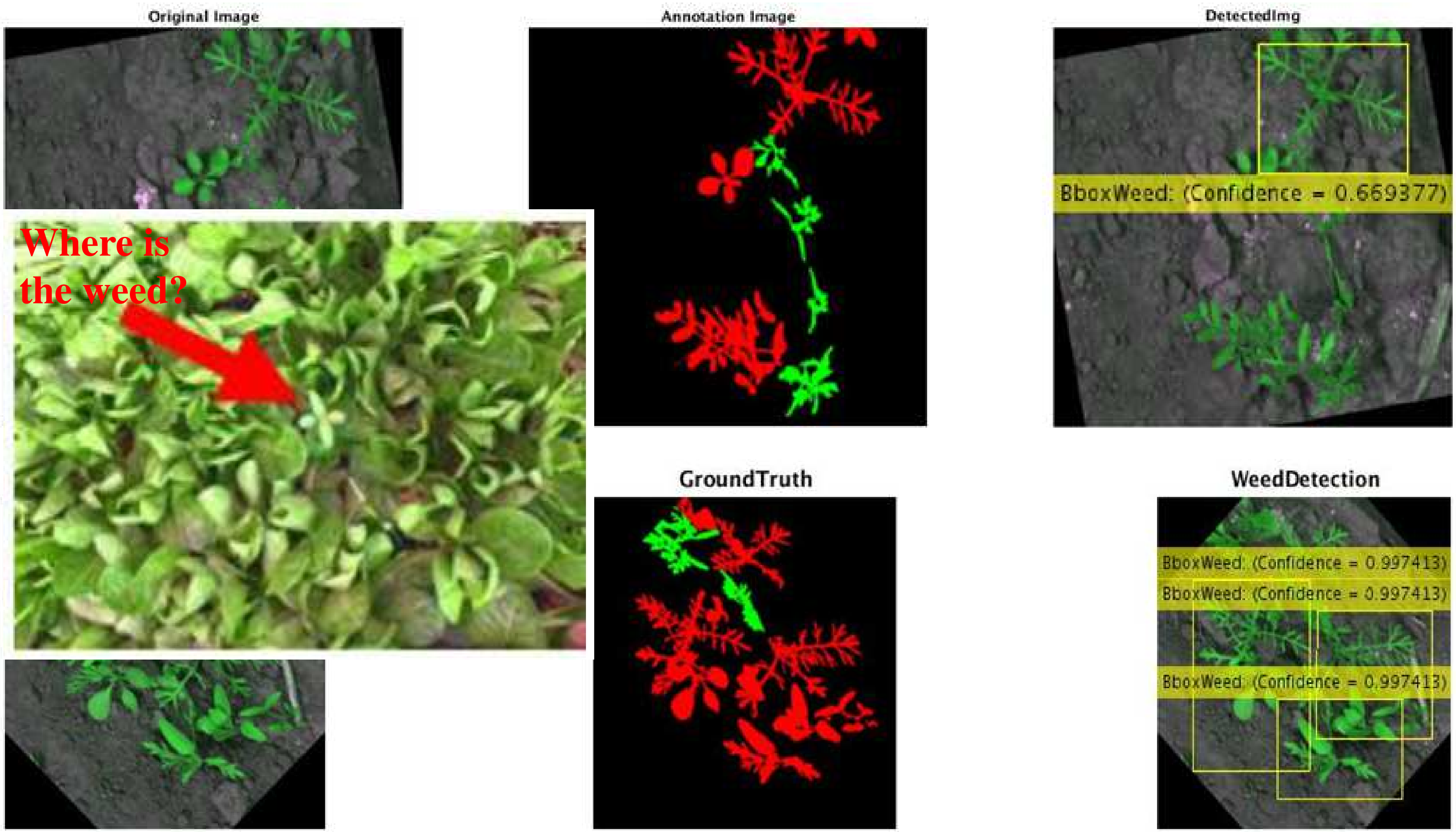
- Prepare ***labelled image***(s) (w/ **labeled regions**) as ***ground truth*** (**GT**) or *Y*
  - Not too difficult, not too expensive

- Train RCNN using Matlab **trainRCNNObjectDetector()** function
  - Augment label data w/ **'PositiveOverlapRange'** & **'NegativeOverlapRange'**
    - Positive detection = results overlap w/ **GT** boxes by 0.5 to 1.0,
      - Measured by the bounding box intersection over union metric
    - Negative detection = results are those overlap w/ **GT** by 0 to 0.5.

```
% Train an R-CNN object detector. This will take several minutes.
rcnn = trainRCNNObjectDetector(stopSignsData, cifar10Net, ...
          options, 'NegativeOverlapRange', [0,  0.4999], ...
          'PositiveOverlapRange', [0.5,  1])
% https://www.mathworks.com/help/vision/ref/trainrcnnobjectdetector.html
```

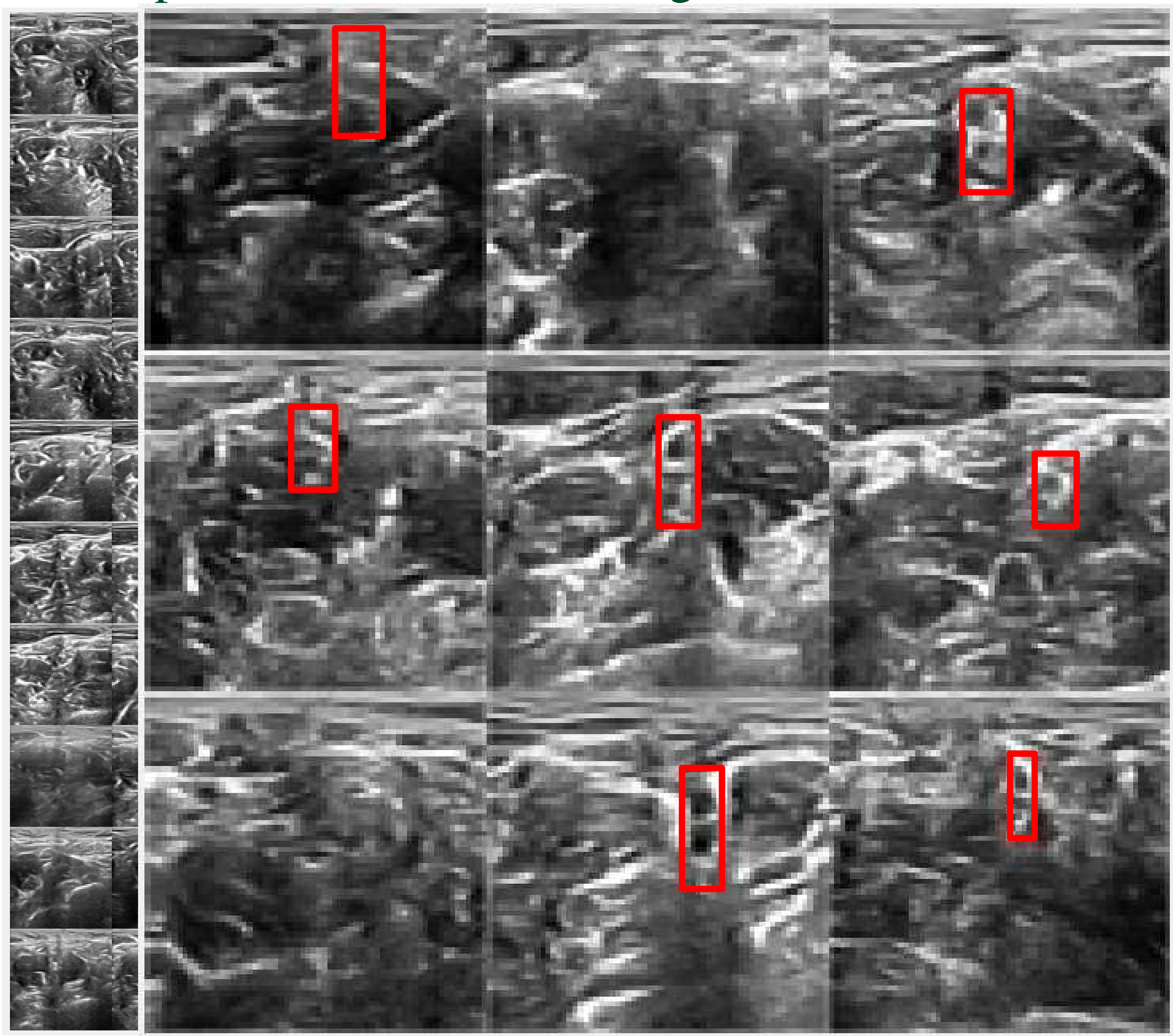
Example, Weed Detection

- Generated by research assistant Karthik Kantipudi, May to June, 2017
- Detection based on **region / bounding-box**

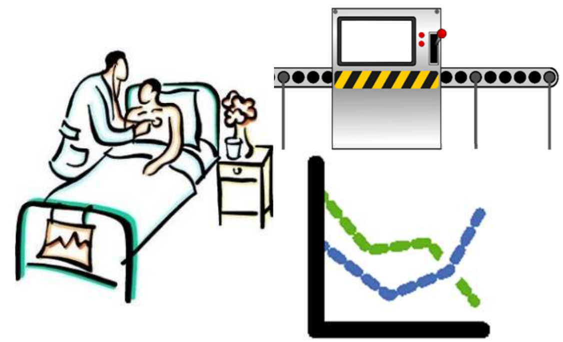


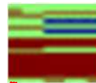
kaggle

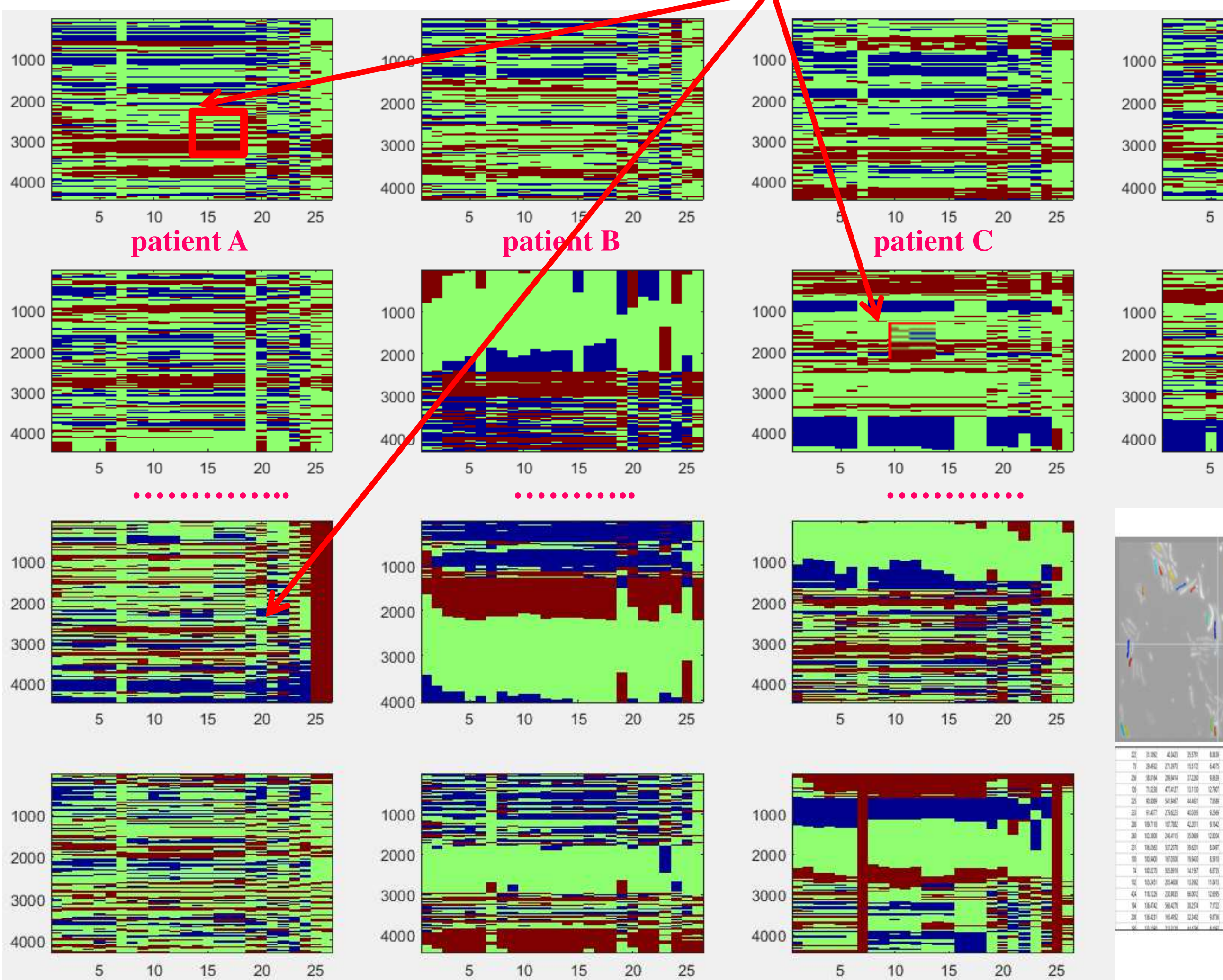
Example, Ultrasound Nerve Segmentation

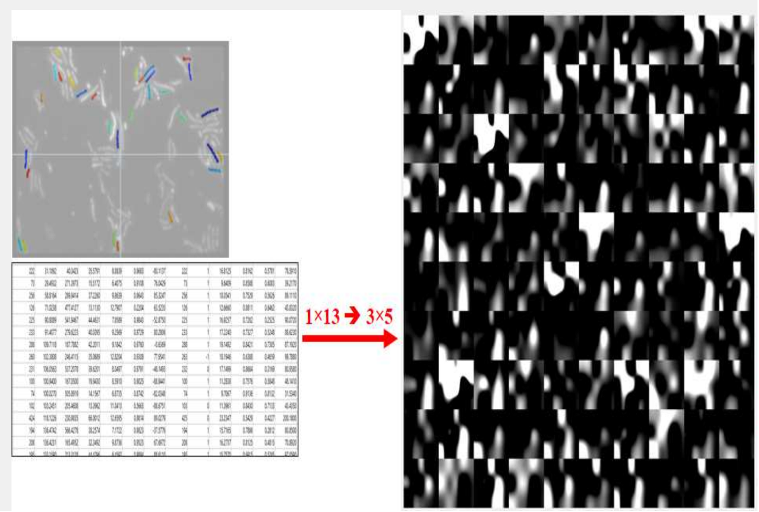


Applying RCNN to Business Data??



- Status of any system, i.e. status of patients over time
  - Any structure? Any pattern?
  - **Find <u>*”similar”*</u> event like this:** 





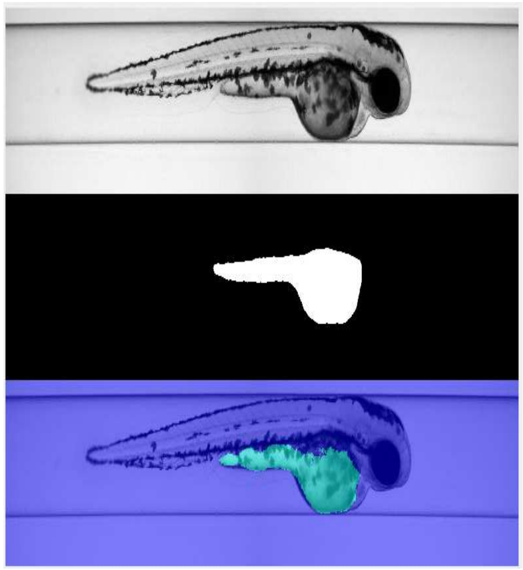
Identifying Organ **PIXELS** in Embryo Images

Yolk

Raw

Labels

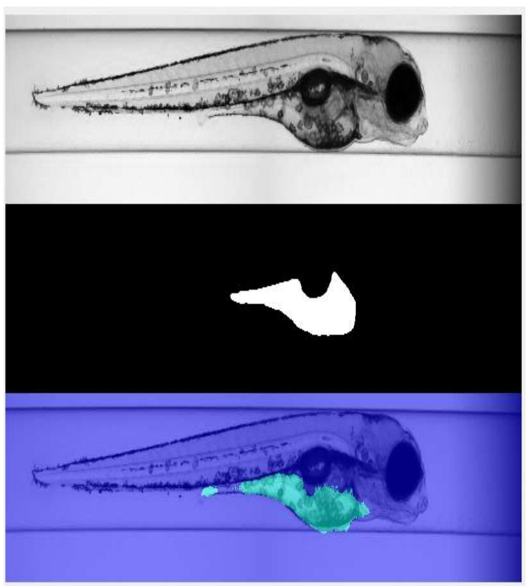
Detection



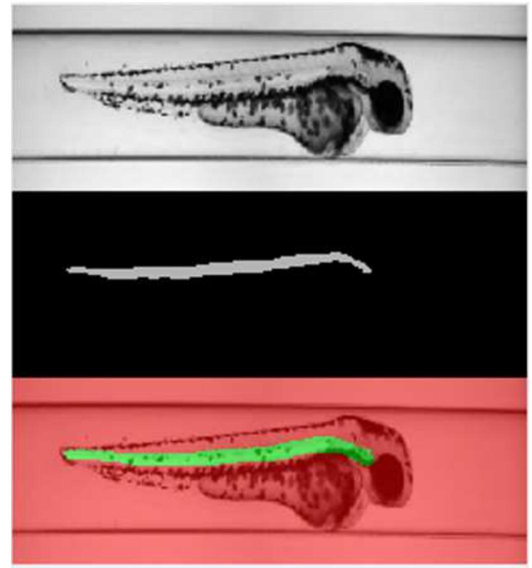
Raw

Labels

Detection



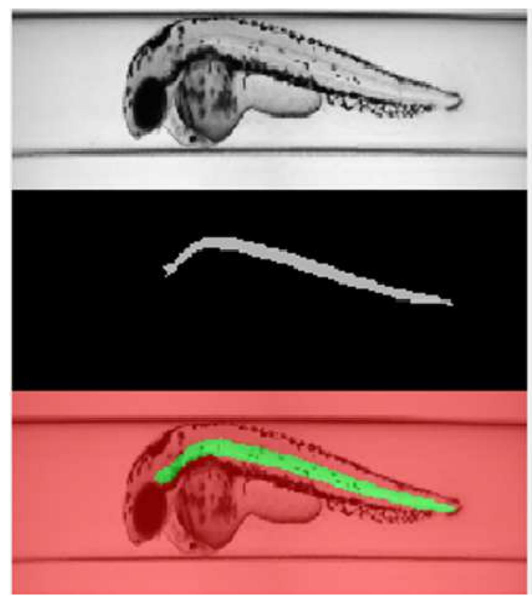
Notochord



Raw

Labels

Detection



Raw

Labels

Detection

NOT

just

finding

bounding

boxes

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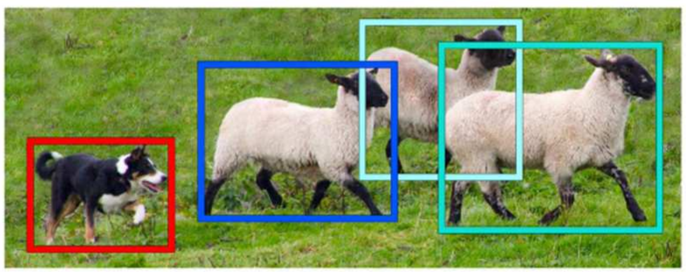

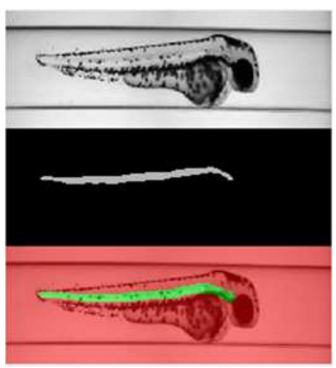

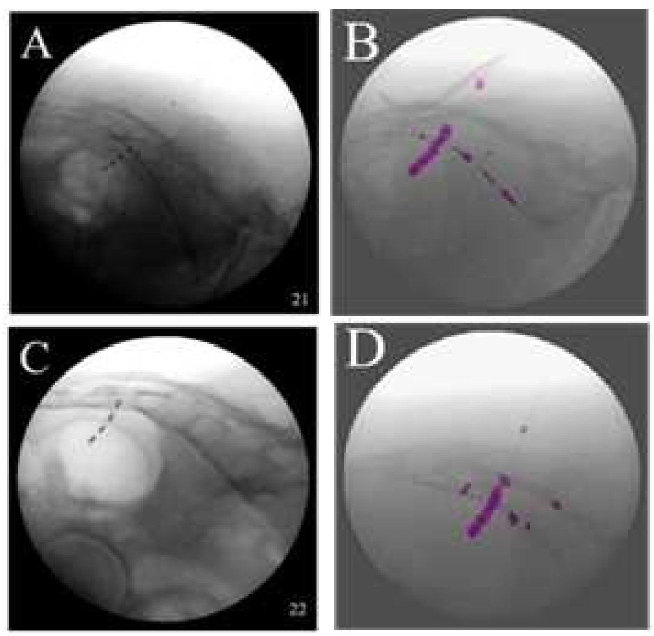

https://youtu.be/hxczTe9U8jY